

Abstraction as Stabilized Replay: Memory, Equivalence, and Semantic Depth

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Abstract

Much contemporary work in artificial intelligence and cognitive theory conflates abstraction with compression, and understanding with representation. This paper argues that such confluations obscure a deeper structural distinction: compression reduces construction cost, while abstraction stabilizes semantic effects under variation. We develop a memory-first framework in which meaning is grounded in causally ordered events and abstraction emerges through equivalence under deterministic replay. On this view, reasoning is not symbolic manipulation but controlled reconstruction, and understanding corresponds to the stabilization of semantic outcomes across counterfactual histories. The framework provides a substrate-neutral account of abstraction that operates below language, explains the limits of purely representational systems, and clarifies the structural conditions under which intelligence-like behavior becomes possible.

1 Introduction

The history of intelligence research is marked by a recurring ambiguity concerning the nature of abstraction. At various times, abstraction has been identified with compression, with generalization, with symbol manipulation, or with representational economy. Each of these notions captures an important aspect of intelligent behavior, yet none adequately distinguishes abstraction from mere data reduction or memorization.

This ambiguity has practical consequences. Systems that excel at compressing large corpora of experience often fail to exhibit understanding, while systems that manipulate abstract symbols frequently lack grounding. These failures suggest that abstraction is neither reducible to compression nor fully captured by representation.

This paper advances a structural account of abstraction grounded in memory and replay. The central claim is that abstraction consists in the stabilization of semantic effects under variation, achieved through equivalence relations induced by construction history. Compression reduces the number of events required to build a structure; abstraction reduces the number of distinct representatives required to produce a given semantic outcome. The two operations are orthogonal.

To articulate this distinction, we adopt a memory-first ontology in which events precede representations, and reasoning consists in deterministic reconstruction rather than symbolic inference. This perspective reframes intelligence as a property of organized construction rather than expressive output.

2 Events, Time, and the Ontology of Memory

Any theory of abstraction presupposes an ontology of structure. In representational accounts, this ontology is typically static: symbols, vectors, or propositions are taken as given, and abstraction is defined over them. By contrast, a memory-first ontology treats structure as something that comes into being through time.

Definition 2.1 (Event). *An event is an irreversible commitment to semantic structure occurring at a unique position in a causal order.*

Events are not observations or descriptions; they are constructive acts. The semantic state of a system at any moment is therefore not a snapshot in isolation, but the cumulative result of all prior events. Memory, on this view, is not a storage medium but a causal artifact.

Time plays a constitutive role. Because events are ordered, semantic state is inherently temporal. Any account of meaning that ignores construction history necessarily discards information relevant to abstraction.

3 Replay as a Model of Reasoning

If semantic state is defined by event history, then reasoning cannot consist merely in forward execution. Instead, reasoning must involve the ability to reconstruct, inspect, and vary construction histories.

Definition 3.1 (Replay). *Replay is the deterministic reconstruction of semantic state from an initial condition and an ordered sequence of events.*

Replay is not simulation in the probabilistic sense; it is exact reconstruction. Its determinism guarantees that identical histories yield identical states, while its inspectability allows semantic consequences to be traced to their origins.

This conception of reasoning differs fundamentally from inference over representations. Rather than manipulating symbols according to syntactic rules, the system re-instantiates the very processes that produced meaning. Counterfactual reasoning becomes possible by replaying modified histories, not by hypothesizing over abstract models.

4 Equivalence and the Emergence of Abstraction

Abstraction arises when multiple distinct construction paths are treated as semantically interchangeable. This interchangeability is formalized through equivalence relations over constructed entities.

Definition 4.1 (Semantic Equivalence). *Two entities are semantically equivalent if, under all admissible continuations of the event history, they induce identical semantic effects.*

Equivalence does not erase history; it constrains future interpretation. When an equivalence relation is introduced, the system commits to treating certain distinctions as irrelevant for subsequent semantic purposes, while retaining the ability to recover their origins through replay.

Proposition 4.1. *Equivalence induces abstraction by reducing the number of representatives required to realize a class of semantic effects.*

This reduction is the essence of abstraction. Importantly, it operates independently of compression. One may introduce equivalence without shortening the event log, and one may compress construction without introducing new equivalences.

5 Compression and Its Limits

Compression reduces description length or construction cost. In event-based systems, this corresponds to reducing the number of events required to reach a given state. Compression is undeniably valuable: it enables efficiency, scalability, and tractability.

However, compression alone does not yield abstraction. A system may compress its history aggressively while preserving every semantic distinction, resulting in a compact but brittle structure. Conversely, a system may retain a long construction history while stabilizing high-level equivalences that enable robust generalization.

This distinction clarifies why systems trained solely to compress observed data often fail to exhibit understanding. They reduce surface complexity without stabilizing semantic invariants.

6 Stability Under Variation

The hallmark of abstraction is stability. An abstract concept is one whose semantic consequences remain invariant under a range of variations in construction history or presentation.

Definition 6.1 (Semantic Stability). *A semantic effect is stable if it is preserved under a nontrivial class of perturbations to the event history.*

Stability is not absolute invariance; it is invariance relative to a class of admissible variations. These variations may include reordering of independent events, substitution of equivalent substructures, or changes in representational form.

Theorem 6.1. *Abstraction corresponds to the identification of stable semantic effects under replay.*

Sketch. Replay deterministically maps event histories to semantic states. If multiple histories map to states that induce identical semantic effects, an equivalence class over those histories can be formed. Stabilizing this equivalence yields abstraction. $\square$

7 Memory, Explanation, and Non-Interference

A memory-first framework distinguishes sharply between causal structure and explanation. Explanation is a projection from memory, not an input to it. As a result, explanations may vary without altering semantic truth.

This separation resolves a fundamental tension in intelligent systems: the need to generate explanations without corrupting reasoning. When explanation feeds back into state, systems become

susceptible to confabulation and self-deception. When explanation is derived but causally inert, understanding remains grounded.

8 Why Representation Is Insufficient

Representational systems operate on summaries of construction rather than on construction itself. They manipulate compressed artifacts whose semantic grounding lies elsewhere. While such systems may approximate abstraction behaviorally, they lack intrinsic access to the conditions under which their abstractions remain valid.

By contrast, a replay-based system can test the stability of its abstractions by varying construction histories directly. It can determine not merely whether a pattern recurs, but whether it survives perturbation.

This capacity marks a qualitative difference between systems that recognize regularities and systems that understand structure.

9 Semantic Depth and Generalization

Abstractions differ not only in scope but in depth. Shallow abstractions capture surface regularities; deep abstractions capture constraints that govern wide classes of constructions.

Definition 9.1 (Semantic Depth). *The semantic depth of an abstraction is the measure of how many independent variations it remains invariant under.*

Depth is revealed through replay: the more variations a semantic effect survives, the deeper the abstraction. This notion aligns generalization with structural robustness rather than statistical frequency.

10 Implications for Intelligence

On the account developed here, intelligence is not the accumulation of representations nor the optimization of predictions. It is the capacity to construct, stabilize, and reuse abstractions grounded in memory.

Systems exhibiting intelligence-like behavior do so because they maintain structures that support replay, equivalence, and counterfactual variation. Systems lacking these properties may achieve impressive performance yet remain fragile, opaque, or ungrounded.

This framework suggests that intelligence emerges not from scale alone, but from the organization of memory and the disciplined management of abstraction.

11 Conclusion

This paper has argued that abstraction is neither synonymous with compression nor reducible to representation. Instead, abstraction emerges from the stabilization of semantic effects under replay, grounded in a memory-first ontology of events.

By distinguishing compression from abstraction, and reasoning from narration, we obtain a framework capable of explaining both the successes and limitations of contemporary intelligent systems. More importantly, we recover a structural notion of understanding that operates below language and representation, offering a foundation for future systems that reason by construction rather than description.